

Ryan Kaddis¹, Alec Lipinski², Chan-Jin Chung¹, Sean Murtha³, Hao Jiang²

¹Dept. of Computer Science, CoAS; ²Dept. of Biomedical Engineering, CoE; ³Orrum Clinical Analytics, Inc.

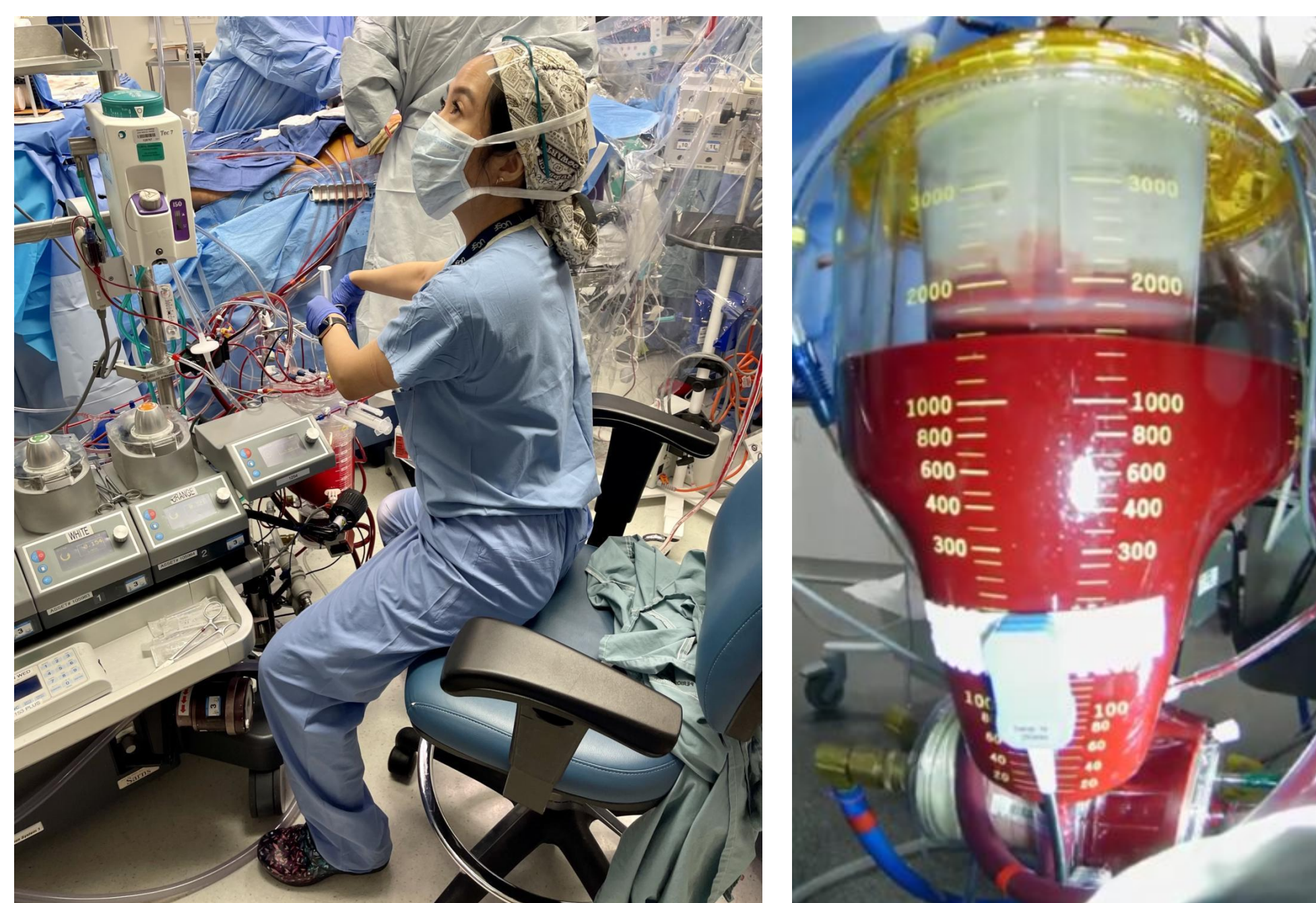
Emails: cchung@ltu.edu; hjiang@ltu.edu

Introduction

Clinical Context & Problem

- Cardiopulmonary Bypass (CPB)** in cardiac surgery requires venous blood storage/monitoring
- Blood volume must be monitored to prevent
 - Air embolism, cerebral injury (volume too high/low)
 - Organ damage (rapid volume changes)
- Perfusionists **manually** read reservoir volume
 - Inefficient, error prone, not practical for data logging
 - Existing sensors only trigger for critical levels

Fig. 1: Operation of Heart-Lung Machine

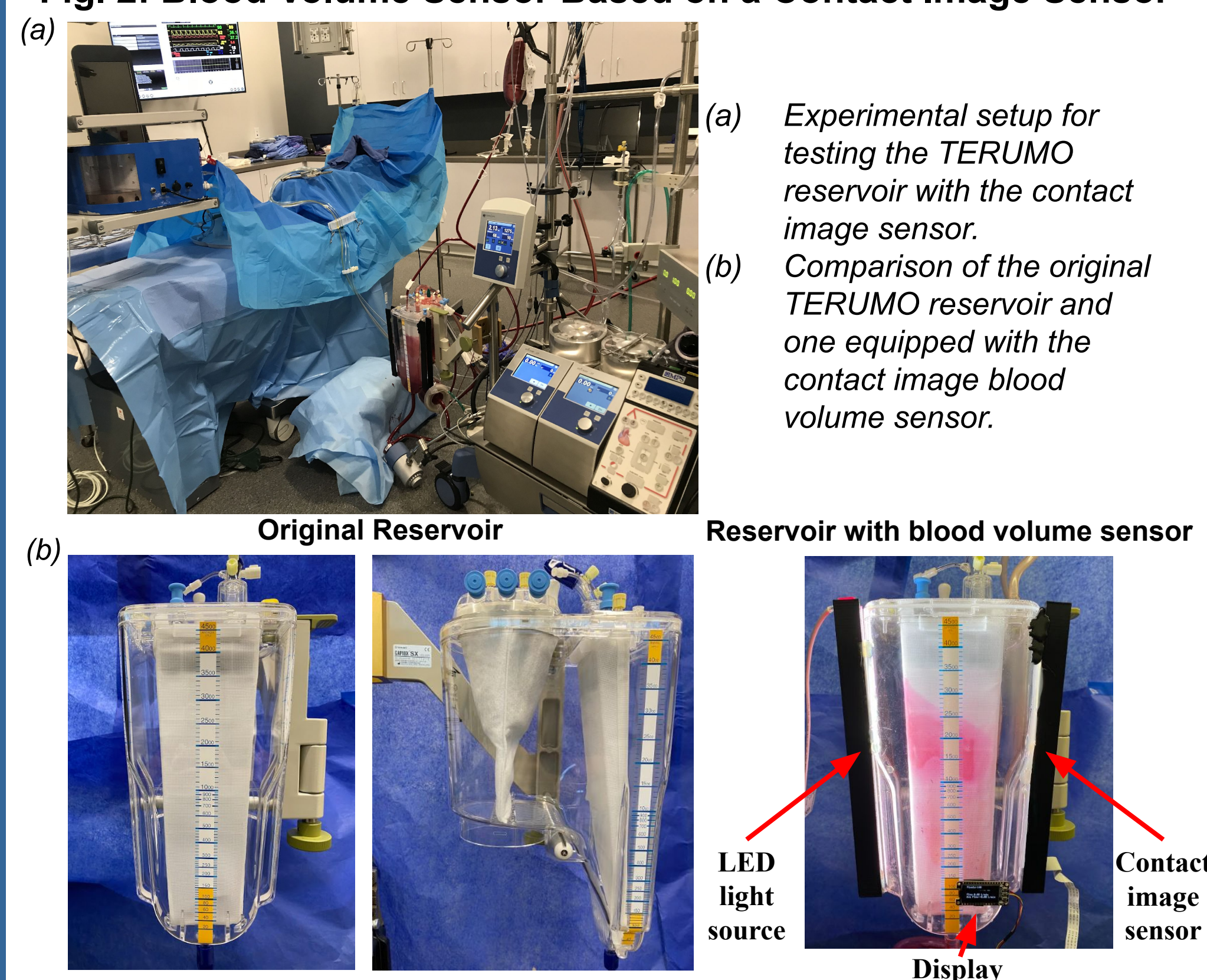


Left: Perfusionist operating heart-lung machine. Right: Venous blood reservoir volume labels.

Previous Solutions

- Optical Sensor – Scans blood level using Contact Image Sensor
 - Accurate, precise, non-intrusive
 - Blocks reservoir visibility
- Gravimetric sensor – Measures volume via weight
 - Accurate, precise, un-obstructive
 - Requires zero-point calibration, affected by movement

Fig. 2: Blood Volume Sensor Based on a Contact Image Sensor^[2]



Objectives

Venous Reservoir Blood Volume Sensor

- Mimics perfusionists' blood volume interpretation
- Convenient for use with clinical CPB
- Accurate (Close to True Value) and precise (Consistent)
- Non-intrusive: no contact with blood
- Non-obstructive: does not block perfusionists' view of the reservoir
- Versatile: compatible with all major brands and models of reservoir
- Advanced functionality: automatic calibration; resistance to mechanical disturbances; data logging

Materials & Methods

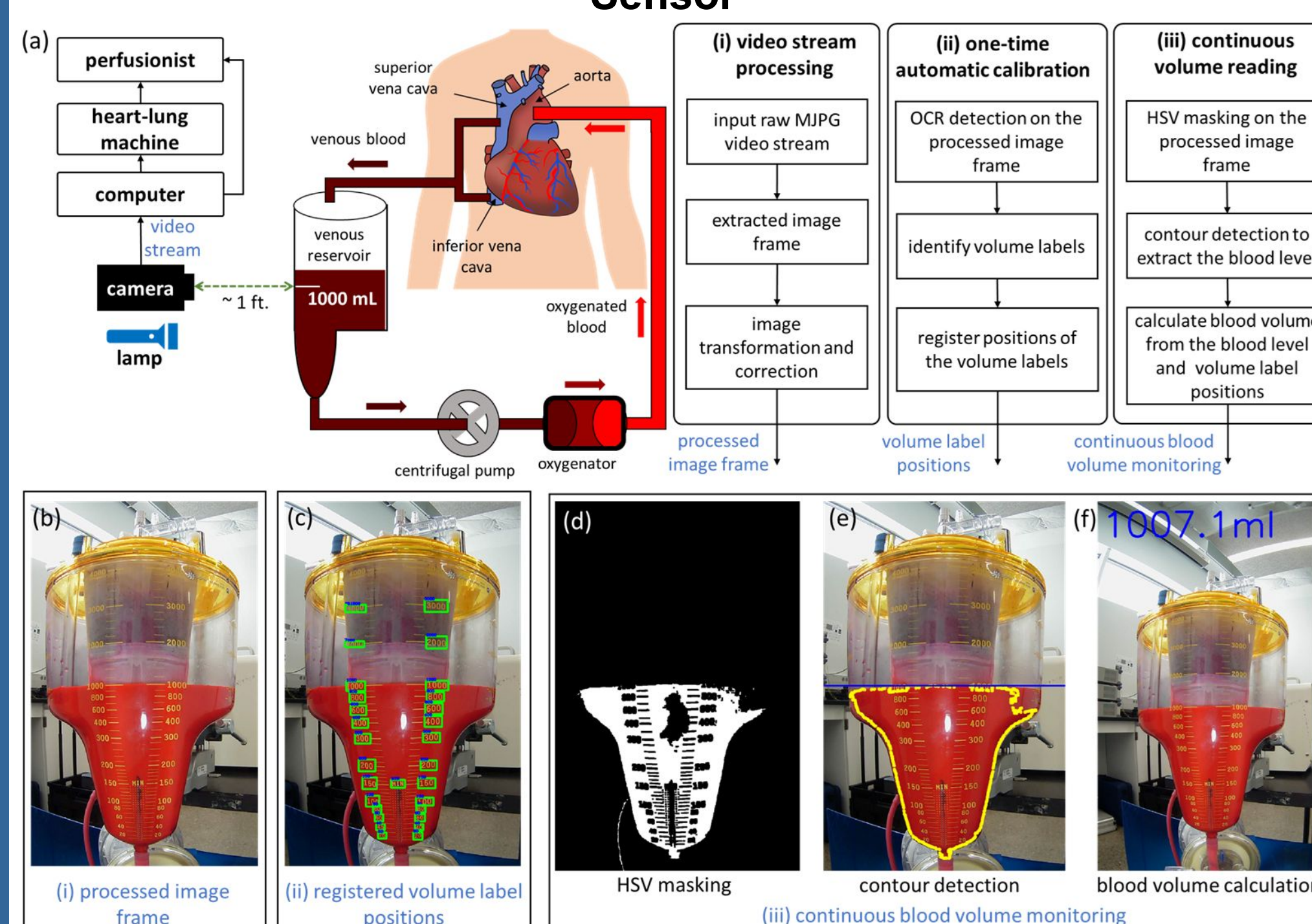
Equipment

- Sensor System:
 - HD High Dynamic Range Webcam – Arducam IMX291
 - LED Lamp with Diffusing Layer
 - Computer – Lenovo ThinkPad P1 Gen 5 Laptop, equipped with NVIDIA RTX A2000 GPU
- Adjustable Arm – Mounts Camera & LED Lamp
- Venous Reservoirs – SORIN INSPIRE HVR & TERUMO CAPIOX NX19
- Test Liquids – Dye/Milk Mixture & Bovine Blood

How Does it Work?

- Computer vision algorithm automatically detects blood volume
- Optical Character Recognition (OCR) locates volume labels
- Contour detection to find the top edge of the blood-colored region

Fig. 3: Algorithm for Computer Vision-based Blood Volume Sensor^[1]



(a) Flowchart for the sensor's functionality. (b) A processed image frame acquired from video stream. (c) One-time automatic calibration based on OCR detection. (d) HSV masking and (e) contour detection used in extracting the blood level. (f) Output image with the calculated blood volume

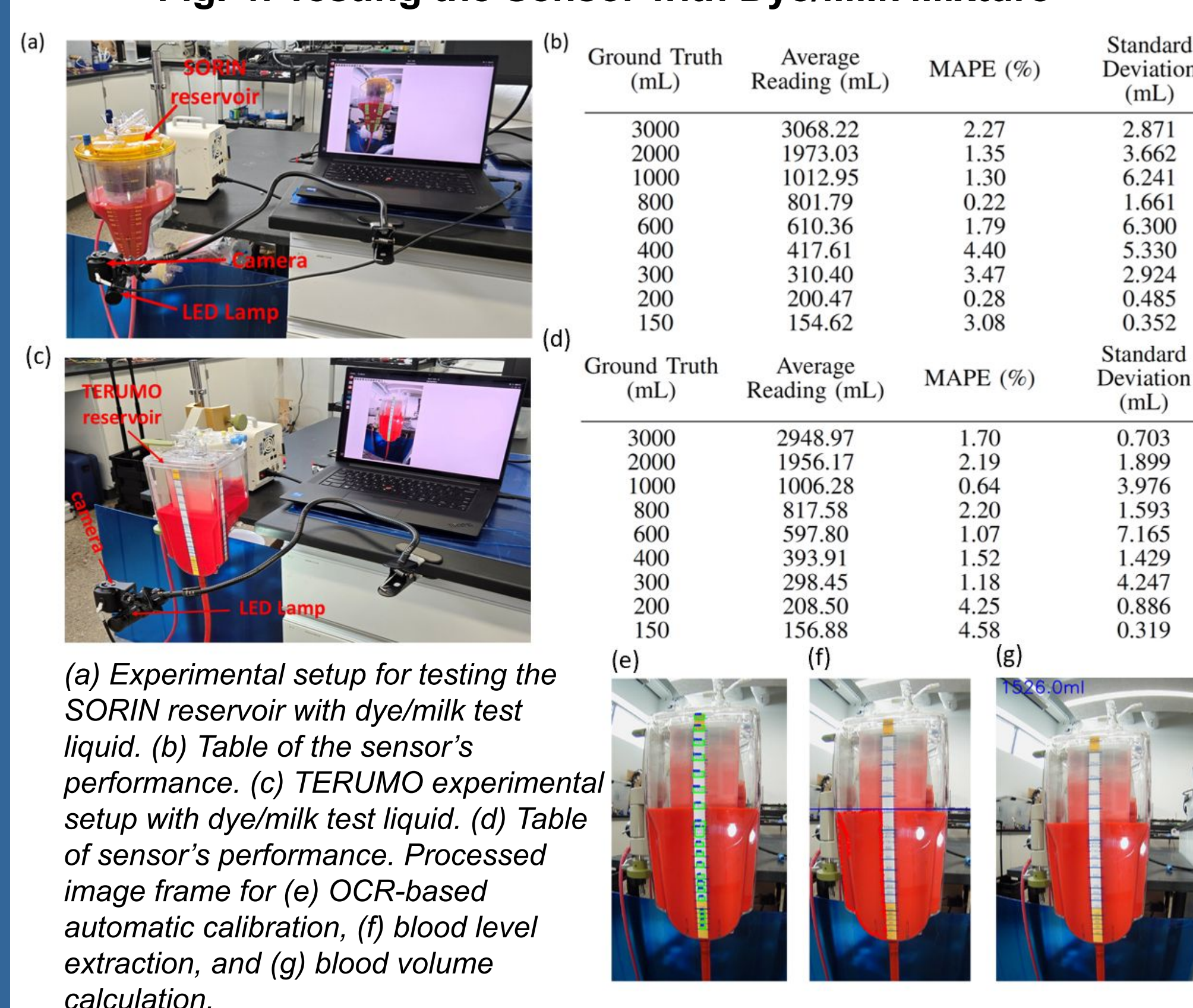
Data Acquisition Protocols

- Fill reservoir with test liquid – Dye/Milk Mixture or Bovine Blood
- Liquid level set to major grid line – e.g. 3000 mL
- Grid Line = "Ground Truth" Volume
- Compare "Ground Truth" Vs. Sensor Reading
- Accuracy: **Mean Absolute Percentage Error (MAPE)**
- Precision: **Standard Deviation**

Experimental Results

- Tested sensor response to SORIN and TERUMO reservoirs using dye/milk mixture
- Recorded MAPE and standard deviation across full volume range without critical-level sensor

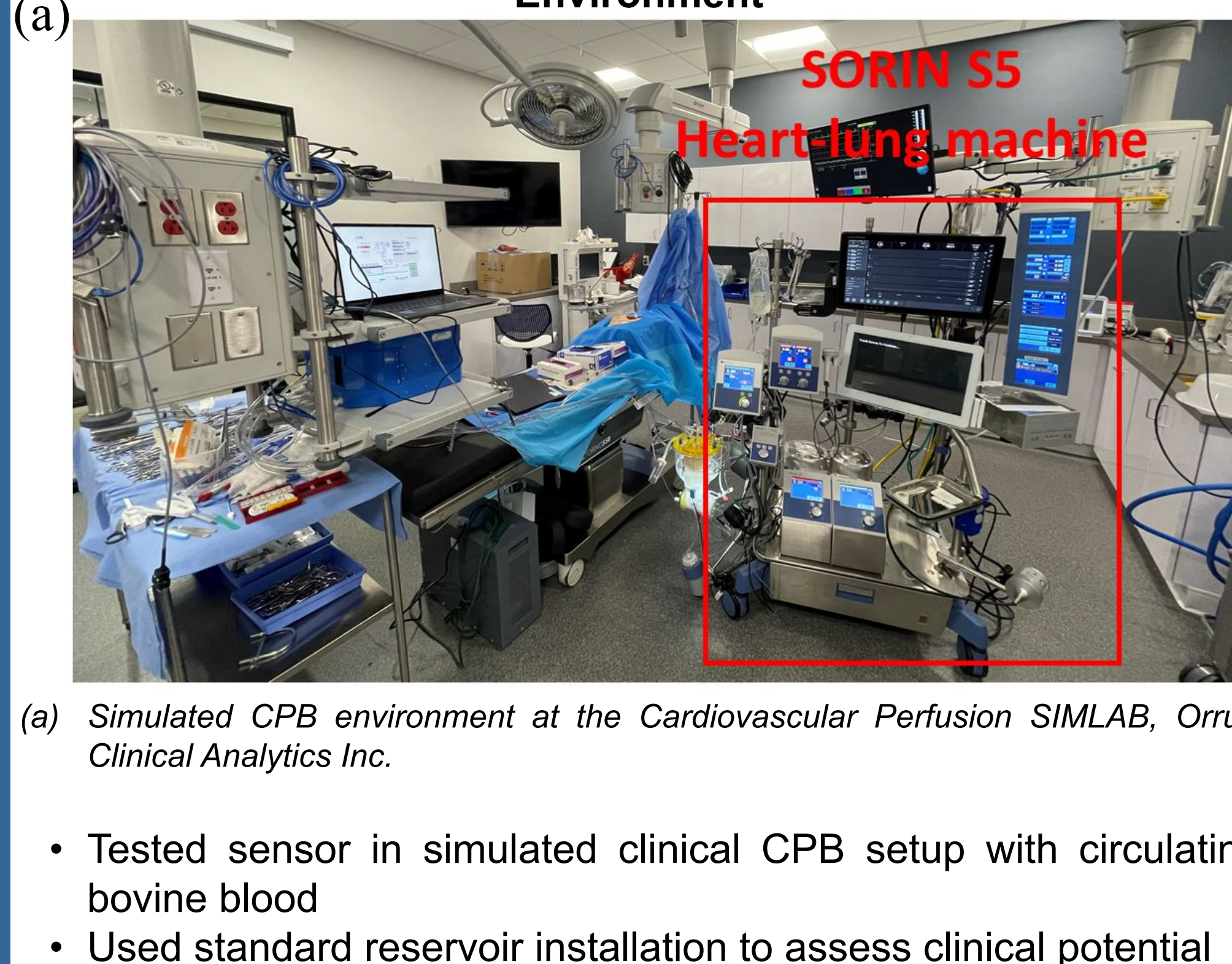
Fig. 4: Testing the Sensor with Dye/Milk Mixture^[1]



Results from In-Lab Testing

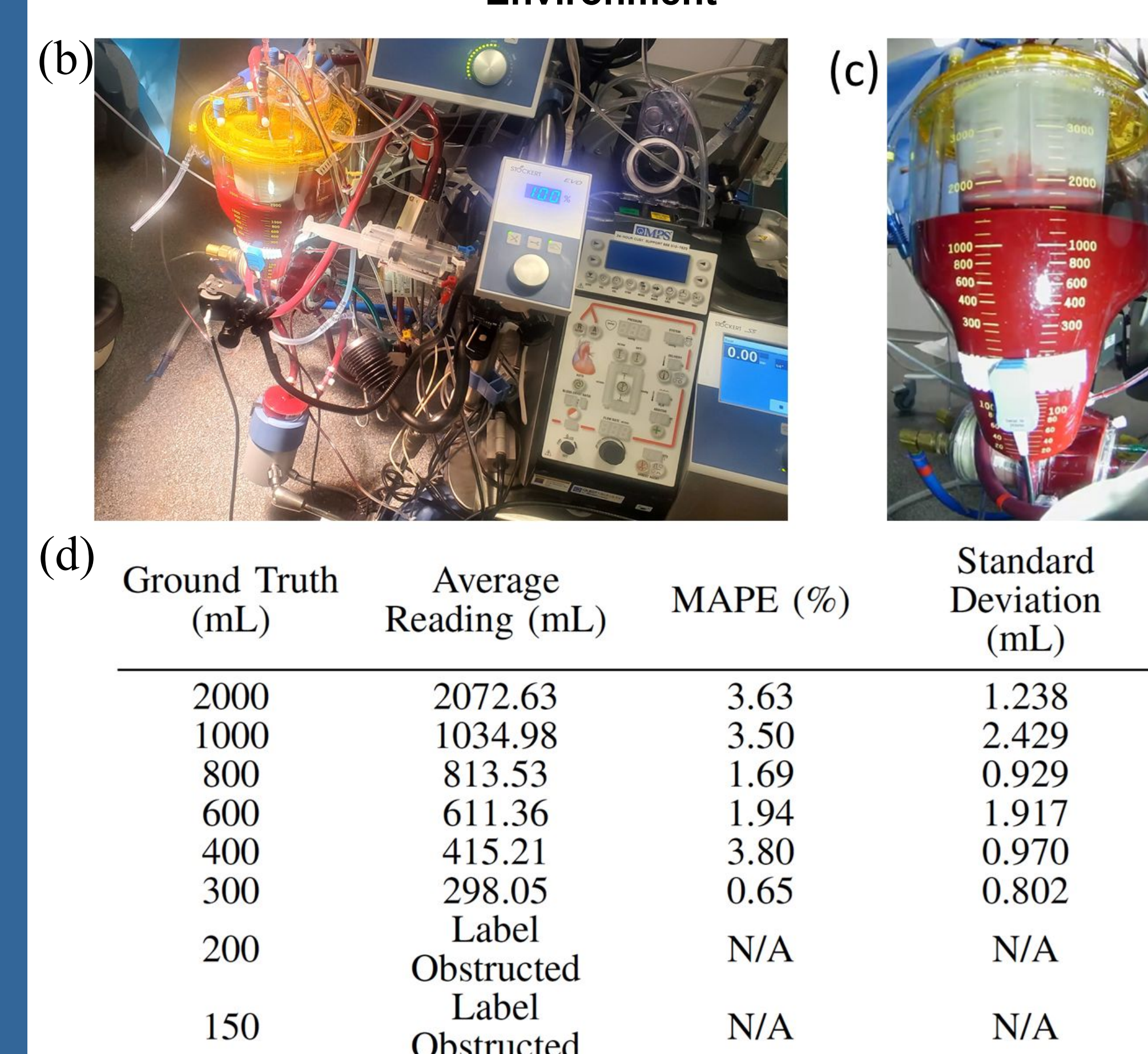
- MAPE
 - SORIN < 2.27%, TERUMO < 2.2% for volumes less than 500 mL
 - MAPE increases to ~4% for volumes below 500 mL
 - Average MAPE across all tests: ~2.5%
- Precision
 - Sensor Precision: 0.3 mL ~ 7 mL
 - Higher Precision below 500 mL due to finer grid line spacing

Fig. 5.1: Testing the Sensor in Simulated Clinical CPB Environment^[1]



- Tested sensor in simulated clinical CPB setup with circulating bovine blood
- Used standard reservoir installation to assess clinical potential

Fig. 5.2: Testing the Sensor in a Simulated Clinical CPB Environment^[1]



(b) SORIN venous reservoir connected to SORIN S5 heart-lung machine. (c) Photo of the reservoir equipped with a critical level safety sensor. (d) Table of the sensor's performance.

Results from Testing in the Simulated CPB Environment

- Critical-level sensor blocked 150 mL and 200 mL labels, restricting the range of our sensor
 - No risk introduced – critical-level sensor stops flow at 300 mL
- Accurate and precise volume readings for all volumes > 300 mL
- Certified perfusionists confirmed the sensor does not obstruct their view
- The system meets clinical CPB requirements without hindering perfusionists' operation

Acknowledgements

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- Special thanks to Andrew Ulaszek for assisting 3D printing and ordering materials.
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Journal & Patent Publications

- [1] R. Kaddis, C.J. Chung, S. Murtha and H. Jiang, "A Non-intrusive, Non-obstructive, Versatile Venous Reservoir Blood Volume Sensor Based on Computer Vision for Clinical Cardiopulmonary Bypass," *IEEE Sensors Letters*, vol. 9, no. 5, pp. 1-4, May 2025, Art no. 6004604
- [2] T. Foley, A. F.-R. i Sabala, M. Fockler, Z. Alzayer, S. Murtha, and H. Jiang, "Continuous contactless measurement of blood volume inside venous reservoirs for cardiopulmonary bypass," *IEEE Sensors Journal*, vol. 23, no. 13, pp. 14 882– 14 890, 2023.
- [3] S. Murtha, H. Jiang, T. Foley, A. F.-R. i Sabala, M. Fockler, Z. Alzayer, and A. Wayne, "Blood volume sensor system," *PCT Patent Application*, p. WO/2023/107591A1, 2023. [Patent pending]
- [4] S. Murtha, R. Kaddis, C.J. Chung, and H. Jiang, "Venous reservoir blood volume sensor with computer vision", *U.S. Provisional Patent Application* No. 63/707,913, filed on Oct 16. [Patent pending]